

# ECD PROJECT

## AIM:

To make a motor driver which can be used to control speed and direction of a motor using a joystick

## EQUIPMENT:

- Motor Driver (Made by using BJTs)
- Analog joystick
- My daq and labview software (control unit)

A motor driver is the cornerstone of robotics as all movement is carried out by motors and motor direction and speed control is done using these

A motor driver consists of a switcher unit and an H bridge

The H bridge is made by using a pnp and npn transistors cascaded with the motor as a load, it essentially diverts power based on which switches are on, thus controlling direction

The speed is controlled by controlling the duty cycle of the pwm that is supplied to the transistor switches switching them on and off with a particular duty cycle (Sign magnitude H bridge)

The analog joystick is to help decide direction and speed of the motor

The software controls and generates pwm using the inbuilt knobs

The myDAQ is going to be used for variable PWM generation which will be sent to the motor driver where a switcher circuit is used to help switch the transistors on and off depending on the signals

The myDAQ can also be used for analog inputs the motors are to be controlled using an external joystick

The Motor Driver is to be externally powered and can be battery powered as well, so the analog ground of the myDAQ is to be connected to the motor driver to avoid wrong values